

Note 6.4 Areas of Surfaces of Revolution

Defining Surface Area

- **Definition**

If the function $f(x) \geq 0$ is continuous and differentiable on $[a, b]$, then the area of the surface generated by revolving the graph $y = f(x)$ about the x -axis is

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$

- **Proof**

For the nonnegative continuous function $y = f(x)$, $a \leq x \leq b$, revolves about the x -axis, we partition the closed interval $[a, b]$ in the usual way. Then the sketched surface approximates as a rectangular. For every small rectangular, its weight is $W = \sqrt{1 + f'(c_k)^2} \cdot \Delta x_k$ and its length is $L = 2\pi f(c_k)$, where c_k satisfies that $\Delta y_k = f'(c_k)\Delta x_k$. So the area of the small rectangular is

$$S_k = 2\pi f(c_k) \sqrt{1 + f'(c_k)^2} \Delta x_k$$

So the sum of the area is

$$S = \sum_{k=1}^n 2\pi f(c_k) \sqrt{1 + f'(c_k)^2} \Delta x_k = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$

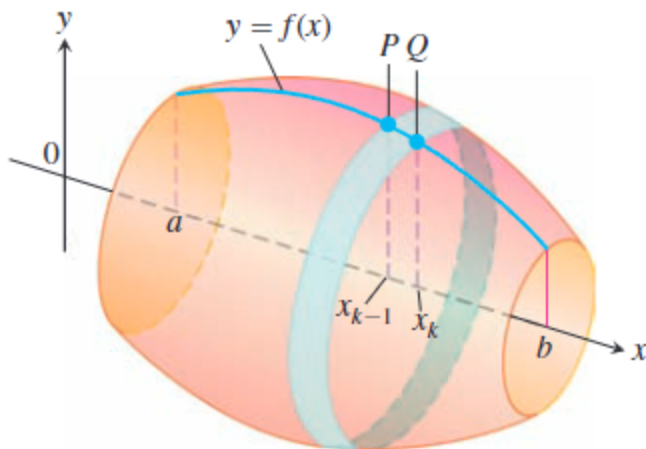


FIGURE 6.30 The surface generated by revolving the graph of a nonnegative function $y = f(x)$, $a \leq x \leq b$, about the x -axis. The surface is a union of bands like the one swept out by the arc PQ .

Revolution About the y -Axis

- Surface Area for Revolution About the y -Axis**

If the function $x = g(y) \geq 0$ is continuous and differentiable on $[c, d]$, then the area of the surface generated by revolving the graph $x = g(y)$ about the y -axis is

$$S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_c^d 2\pi g(y) \sqrt{1 + (g'(y))^2} dy$$

Exercise Section 6.4